

Clinical Lab Analysis System — Test Report

Test Report

Automated tests for the Clinical Lab Analysis System.

Summary

- **170 tests pass, 0 failures.**
- **94% line coverage** (target: 85%).
- Layers covered: unit tests for every logic module (including the code-gen sandbox, programmer agent, and orchestration loop) + an end-to-end test for the Streamlit UI that enters both role workspaces (Doctor / Researcher) and exercises **all 12 questions** and the experimental code-gen mode.

Run them with:

```
uv sync
uv run pytest
```

(Coverage is configured in `pyproject.toml` with an 85% minimum gate.)

Coverage

Module	Coverage
agent_pipeline.py	100%
code_generation.py	100%
correction.py	100%
execution.py	100%
llm_router.py	100%
result_presentation.py	100%
sandbox.py	99%
data_prep.py	96%
main_pipeline.py	95%
validation.py	95%
app.py	94%
code_agent.py	94%
presentation_agent.py	92%
question_router.py	91%

Module	Coverage
questions.py	91%
llm_client.py	90%
Total	94%

End-to-end results per question

Each question is selected in the app, the correct parameter widgets are asserted to render, the analysis is run, and the result components are checked. These were verified both via Streamlit's `AppTest` harness (`tests/test_e2e_app.py`) and in a real headless Chromium browser (screenshots in `docs/assets/screenshots/`).

#	Question	Widgets shown	Expected result	Status
1	Abnormal lab results	Admission	Table, all <code>FLAG = abnormal</code>	Pass
2	Latest lab value	Admission, Lab Test	Exactly 1 row	Pass
3	Lab trend over time	Admission, Lab Test	Time-sorted table + line chart	Pass
4	All lab tests	Admission	Table of all tests	Pass
5	First vs last value	Admission, Lab Test	1-row comparison incl. difference	Pass
6	Summary statistics	Admission, Lab Test	1 row: count/min/max/mean	Pass
7	Abnormal vs normal counts	Admission	1 row: total/abnormal/normal	Pass
8	Abnormal results for a lab	Admission, Lab Test	Abnormal rows, or graceful "no records"	Pass
9	Distinct abnormal tests	Admission	Table of tests with abnormal counts	Pass
10	Values in a date range	Admission, Lab Test, Date range	Time-sorted table + line chart	Pass
11	All admissions for a patient	Patient ID	Table of the patient's admissions	Pass
12	Abnormal vs normal across all admissions	(none - dataset-wide)	One row per admission: total/abnormal/normal, most abnormal first	Pass

Edge cases covered

- **No-records path:** question 8 with a lab that has no abnormal readings shows the neutral "no matching records" banner instead of an error.
- **Correction loop:** a generated function-name typo is auto-corrected and re-run.
- **Regression guard:** the typo corrector no longer corrupts the valid name `get_abnormal_results_for_lab` (it matches whole identifiers only).
- **Missing data:** data functions return empty frames for unknown admissions/patients.
- **Validation failures:** execution error, `None` result, non-DataFrame, missing columns, empty frame, and failing per-question rules are all handled.

- **Dataset-wide aggregate (Q12):** per-admission counts are cross-checked against the single-admission view (Q7), and `Abnormal + Normal == Total Tests` is validated for every row. The "all admissions" natural-language query routes here (rules and LLM) instead of demanding an admission ID.
- **Code-gen sandbox (experimental mode):** imports, `open`, dunder/underscore attributes, file/network/eval methods (`to_csv` , `query` , ...), loops, and syntax errors are all rejected by the AST allowlist; runtime errors, timeouts, and row caps are handled. The agent loop is tested end-to-end with a fake LLM: it refines after a runtime error, blocks malicious code, and degrades gracefully on LLM errors.

Example (question 11, patient overview)

Deploy

Doctor Workspace

Fast admission-level review for bedside decisions.

Back to Welcome

Switch to Researcher

Doctor focus: Use this view when the user wants to inspect a patient admission, find abnormal values, review trends, or compare lab movement during the stay.

Admission reviewLatest valueTrendAbnormal flags

> Doctor questions

> Ask a clinical lab question in natural language (optional)

Question Selection

CategoryAllChoose a clinical questionList all admissions for a specific patient

All admissions recorded for one patient. Inputs: patient.

Patient ID (SUBJECT_ID)2.0Run Analysis

Selected Question

[11] List all admissions for a specific patient

Selected Parameters

{ "subject_id" : 2 }

Patient Context

Patient 2 has 1 admission(s) in the dataset.

Analysis completed successfully

ExecutionExecution completed successfully

ValidationPassed

Correction AppliedNo

Overall StatusExecution and validation completed successfully

Pipeline Explanation

STEP 1

Question selected

[11] List all admissions for a specific patient. Inputs: patient. Parameters: {'subject_id': 2}.

STEP 2

Code generated

The registry selected 'get_patient_admissions' and produced a Python call from the parameters.

STEP 3

Execution

Execution completed successfully

STEP 4

Validation

Validation passed successfully.

STEP 5

Correction

No correction was needed; the first valid attempt passed.

Returned rows: 1. Attempts made: 1.

Final ResultValidation AuditCorrection AttemptsGenerated Code

Final Result

Rows returned1

SUBJECT_ID	HADM_ID	ADMITTIME	DISCHTIME	DIAGNOSIS
2	163353	2138-07-17 19:04:00	2138-07-21 15:48:00	NEWBORN

Download Result as CSV

Clinical Lab Analysis System — Hala Hillou, Dana Nmarry, Anas Khoury, Alon Engel — Seminar: Software Engineering in the Age of AI, University of Haifa, 2026